Visualizing EM profile data in QGIS

Two files are needed: (1) a CSV with the start and end coordinates of each profile, and (2) a CSV with the EM measurements (VL and HL) at every 5 m station, without coordinates. QGIS calculates the intermediate positions from the line.

Input files

File 1 — profile endpoints (2 rows per profile, one for start, one for end)

profile_id	order	station_m	easting_x	northing_y	crs
Profile 1	1	0	465879	5754916	EPSG:32633
Profile 1	2	140	466010	5754856	EPSG:32633

File 2 — EM measurements (one row per station, no coordinates)

profile_id	station_m	EM_VL_mSm	EM_HL_mSm
Profile 1	0	14.5	15.9
Profile 1	5	21.0	13.9
Profile 1	10	18.7	21.5
Profile 1	...	...	...
Profile 1	140	8.8	29.4

Step 1 — Load the profile endpoints

Go to Layer → Add Layer → Add Delimited Text Layer.

1. Delimiter: Comma
2. Geometry definition: Point coordinates — X field: easting_x, Y field: northing_y
3. CRS: EPSG:32633
4. Add.

Two points appear on the map — start and end of the profile.

Step 2 — Create the profile line

Open the Processing Toolbox (Ctrl+Alt+T) and search for Points to path.

5. Input layer: the 2-point layer
6. Order by field: order
7. Group by field: profile_id
8. Run.

A straight line between the two points appears on the map.

Step 3 — Generate points every 5 m

In the Processing Toolbox search for Points along geometry (under Vector Geometry).

9. Input layer: the line from Step 2
10. Distance: 5 (meters)
11. Run.

29 points are generated (0, 5, 10 ... 140 m), each with a field called distance storing the station value.

Step 4 — Load the EM measurements table

Go to Layer → Add Layer → Add Delimited Text Layer.

12. Select the EM measurements CSV
13. Geometry definition: No geometry
14. Add.

The table loads in the Layers panel with no geometry. This is correct — coordinates are not needed here.

Step 5 — Join measurements to the points

Right-click the 29-point layer → Properties → Joins → click +.

15. Join layer: P1_EM_mediciones
16. Join field: station_m
17. Target field: distance
18. Under Joined fields, select only EM_VL_mSm and EM_HL_mSm
19. OK → Apply.

Open the Attribute Table to check — the EM columns should have values at every point.

Step 6 — Visualize the conductivity values

Right-click the point layer → Properties → Symbology.

20. Change Single symbol to Graduated
21. Column: EM_HL_mSm or EM_VL_mSm
22. Color ramp: RdYlGn reversed, or Spectral
23. Classify → OK.

Step 7 — Compare HL and VL side by side

Duplicate the layer to have one copy for each configuration.

24. Right-click the point layer → Duplicate layer.
25. On the original: Properties → Symbology, set column to EM_HL_mSm, classify, OK.
26. On the duplicate: Properties → Symbology, set column to EM_VL_mSm, classify, OK.
27. Rename each layer — right-click → Rename. For example: P1_HL and P1_VL.
28. Toggle them on and off in the Layers panel to compare.

Both layers share the same point geometry — only the displayed value changes.

HL vs VL — which penetrates more?

The horizontal loop (HL) penetrates deeper. When the coil is horizontal the magnetic dipole is vertical, and the field lines close at greater depth. The vertical loop (VL) is the opposite — the dipole is horizontal and the field stays close to the surface. As a rough rule, HL reaches about twice the depth of VL for the same coil separation.